

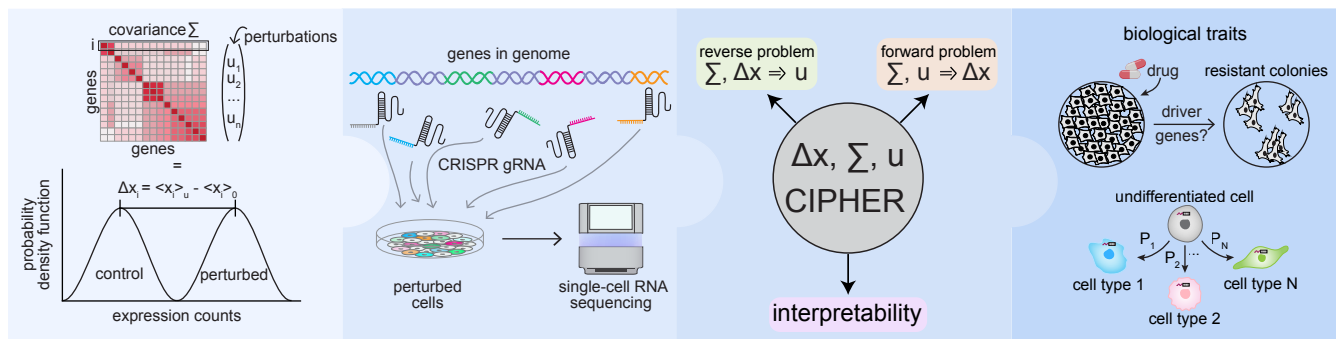


Fig. 1: Schematic of the workflow and applications to synthetic and real experimental datasets. (column 1) The gene-gene covariance matrix informs on changes to gene expression upon perturbation. (column 2) Perturb-seq interrogates thousands of perturbations by combining RNA-seq with CRISPR screens. (column 3) Given covariance and expression changes, CIPHER has three complementary modalities. The forward problem fits changes in expression, the reverse problem predicts the driving perturbation(s) and the framework is fully interpretable by telling us 1) how each genes change is made up of contributions from both itself and every other gene and 2) how response is propagated along dominant or subdominant modes of the covariance. (column 4) CIPHER can be readily applied to biological traits. In cancer drug resistance, CIPHER can identify resistance drivers from the fluctuations in naive cancer cells. In differentiation, CIPHER can infer cell fate decisions from the fluctuations of undifferentiated cells.